Custom RISC-V AI Processor

Project Overview

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Contents

1 Project Description

This project focuses on the design and implementation of a custom 32-bit RISC-V processor integrated with AI-oriented instruction extensions. The processor is developed completely in Verilog RTL and implements a classical five-stage pipelined architecture consisting of:

- Instruction Fetch (IF)
- Instruction Decode (ID)
- Execute (EX)
- Memory Access (MEM)
- Write Back (WB)

In addition to standard RISC-V instructions, the processor introduces custom AI instructions through a dedicated AI execution unit integrated within the execute stage.

2 Project Objectives

- Design a complete 32-bit pipelined RISC-V processor.
- Implement hazard handling mechanisms.
- Develop custom ISA extensions for AI workloads.
- Integrate an AI execution engine into the processor datapath.
- Build a scalable architecture capable of supporting future vector and CNN accelerators.
- Verify functionality through RTL simulation and waveform analysis.

3 Processor Architecture

The processor implements a standard five-stage pipeline:

IF	⟵	ID	⟵	EX	⟵	MEM	⟵	WB

Implemented Modules

- Program Counter
- Instruction Memory
- Register File
- Immediate Generator
- Control Unit
- ALU
- Data Memory
- Pipeline Registers

- Forwarding Unit
- Hazard Detection Unit
- Branch Handling Logic
- AI Execution Unit

4 Instruction Support

4.1 Standard RISC-V Instructions

- ADD
- SUB
- AND
- OR
- ADDI
- LW
- SW
- BEQ

4.2 Custom AI Instructions

Currently implemented:

- AIMAX

Planned AI extensions:

- Vector AI Instructions
- Vector Multiply-Accumulate
- Matrix Multiplication Accelerator
- Tiny CNN Accelerator

5 Pipeline Optimization Features

5.1 Forwarding Unit

Implements data forwarding paths from:

- EX/MEM stage
- MEM/WB stage

to minimize RAW hazards and reduce pipeline stalls.

5.2 Hazard Detection Unit

Handles load-use hazards by:

- Freezing Program Counter
- Freezing IF/ID pipeline register
- Inserting pipeline bubbles

5.3 Branch Infrastructure

Supports:

- Branch comparison logic
- PC redirection
- Control flow management

6 Current Architecture

```
          +-----------------+
          | Instruction Mem|
          +--------+-------+
                   |
                   v

IF --> ID --> EX --> MEM --> WB
                 |
                 |
          +------+------+
          | AI Execution|
          |    Unit      |
          +--------------+
```

7 Future Roadmap

The project is planned to evolve into a complete AI-capable processor.

RISC-V CPU
$\downarrow$
Vector AI Engine
$\downarrow$
Matrix Multiply Accelerator
$\downarrow$
Tiny CNN Accelerator

Future capabilities include:

- SIMD-style vector operations

- Matrix multiplication acceleration
- Hardware convolution engine
- ReLU and pooling accelerators
- Memory-mapped AI accelerator interface

8 Tools Used

- Verilog HDL
- Xilinx Vivado
- RTL Simulation
- Waveform Verification
- Git and GitHub

9 Learning Outcomes

This project provides practical experience in:

- Computer Architecture
- RTL Design
- Pipelined Processor Design
- Hazard Handling
- Custom ISA Extensions
- AI Hardware Acceleration
- FPGA-Oriented Design Methodology

10 Conclusion

The Custom RISC-V AI Processor demonstrates the implementation of a complete pipelined processor integrated with AI-oriented extensions. The architecture serves as a foundation for future accelerator development, including vector engines, matrix processors, and CNN hardware acceleration.

The project combines concepts from processor design, digital systems, computer architecture, and hardware acceleration, making it suitable for RTL, FPGA, ASIC Design, and Processor Architecture applications.